

Experimental fact	Our verdict	Why it follows
Anhydrous (5.5 Å) is not SC (<i>Foo 2003</i>)	✓ explained	$\alpha > 0$: single stiff well ($\hbar\omega \approx 30$ meV), $N(0) \approx 0$ — no 2DEG, nothing to pair.
Monolayer hydrate (6.9 Å) is not SC (<i>Foo 2003</i>)	✓ explained	Past the window: $\lambda \approx 20 \gg 2$, the alkali self-traps (polaronic), so SC is killed.
Bilayer hydrate (9.9 Å) is SC, $T_c = 4.5$ K (<i>Takada 2003</i>)	▲ prediction	Bare geometry is polaronic here; SC needs water to soften the Na well back into the window — untested.
No superconducting Li analogue	✓ explained	$c_{\text{Li}}^* \leq 5.5$ Å and lighter mass $\Rightarrow$ stiffer mode; Li never enters the soft-mode window.
Magnetic phase borders the dome (<i>expt.</i>)	✓ explained	Local moment turns on exactly at the $\alpha < 0$ well transition (Fig. 4), coincident with the dome.
Superconducting T_c is only a few K (<i>Takada 2003</i>)	✓ explained	Allen–Dynes with the DFT $\omega_{\text{eff}}, \lambda$ gives peak $T_c \approx 14$ K (band 11–23) — the right scale.